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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Motion in a Straight Line

JEE Main Physics · Previous Year Questions with Answer Key

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INSIDE THIS PDF

MCQ (Single Correct Answer)	104 questions
Numerical	26 questions
Answer key	at the end

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MCQ (Single Correct Answer) 104 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A gas balloon is going up with a constant velocity of $10\ \mathrm{m/s}$. When this balloon reached a height of 75 m , a stone is dropped from it and balloon keeps moving up with the same velocity. The height of the balloon when the stone hits the ground is ____ m. (Take $g = 10\ \mathrm{m/s^2}$)

Open solution

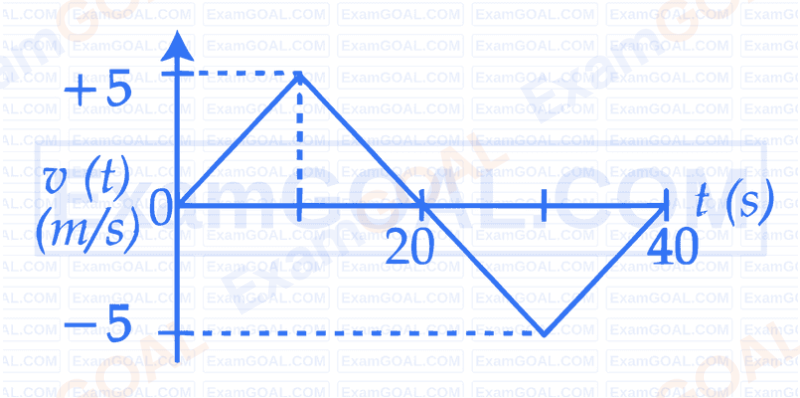
(A) 85 (B) 150 (C) 129 (D) 125

Q2. JEE Main 2026 (Online) 5th April Evening Shift

The velocity (v) versus time (t) plot of a particle is shown in the figure, for a time interval of 40 s . The total distance travelled by the particle and the average velocity during this period are, respectively

Open solution

____.



(A) 25 m and zero (B) 50 m and zero

(C) 100 m and zero (D) 100 m and $2.5\ \mathrm{m/s}$

Q3. JEE Main 2026 (Online) 4th April Evening Shift

Two cars A and B are moving in the same direction along a straight line with speeds $100\ \mathrm{km/h}$ and $80\ \mathrm{km/h}$, respectively such that car A is moving ahead of car B. A person in car B throws a stone with a speed v so that it hits the car A with a speed of $5\ \mathrm{m/s}$. The value of v is ____ $\mathrm{km/h}$.

Open solution

(A) 18 (B) 28 (C) 38 (D) 48

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Q4. JEE Main 2026 (Online) 28th January Evening Shift

A particle starts moving from time $t = 0$ and its coordinate is given as $x(t) = 4t^3 - 3t$

A. The particle returns to its original position (origin) 0.866 units later

B. The particle is 1 unit away from origin at its turning point

C. Acceleration of the particle is non-negative

D. The particle is 0.5 units away from origin at its turning point

E. Particle never turns back as acceleration is non-negative

Choose the **correct** answer from the options given below :

Open solution

(A) A, C, D Only **(B)** A, B, C Only **(C)** C, E Only **(D)** A, C Only

Q5. JEE Main 2026 (Online) 28th January Morning Shift

Water drops fall from a tap on the floor, 5 m below, at regular intervals of time, the first drop strikes the floor when the sixth drop begins to fall. The height at which the fourth drop will be from ground, at the instant when the first drop strikes the ground is ____ m.

$\left(g = 10\ m/s^2\right)$

Open solution

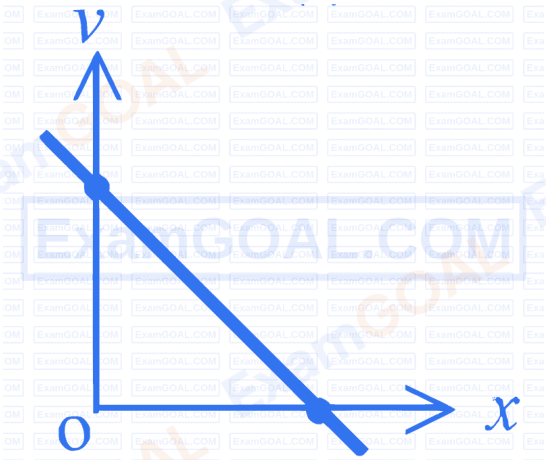
(A) 3.8 **(B)** 4.0 **(C)** 4.2 **(D)** 2.5

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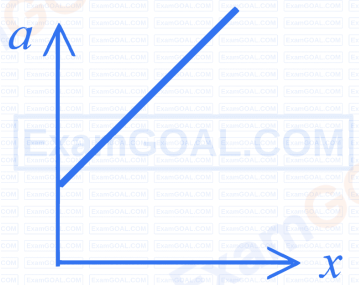
Q6. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

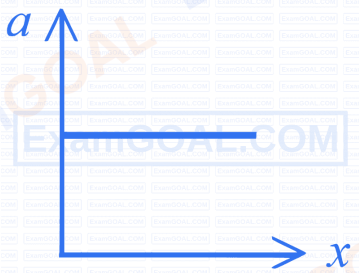
The velocity (v) - Distance (x) graph is shown in figure. Which graph represents acceleration(a) versus distance (x) variation of this system?



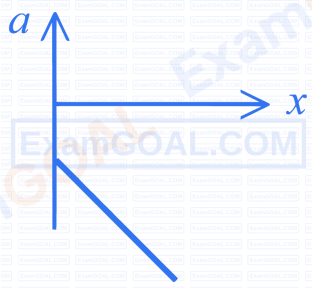
(A)



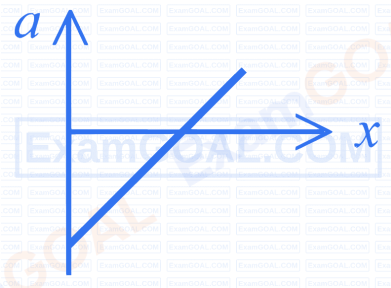
(B)



(C)



(D)



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Q7. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

A paratrooper jumps from an aeroplane and opens a parachute after 2 s of free fall and starts decelerating with $3\ \mathrm{m/s^2}$. At 10 m height from ground, while descending with the help of parachute, the speed of paratrooper is $5\ \mathrm{m/s}$. The initial height of the airplane is ____ m.

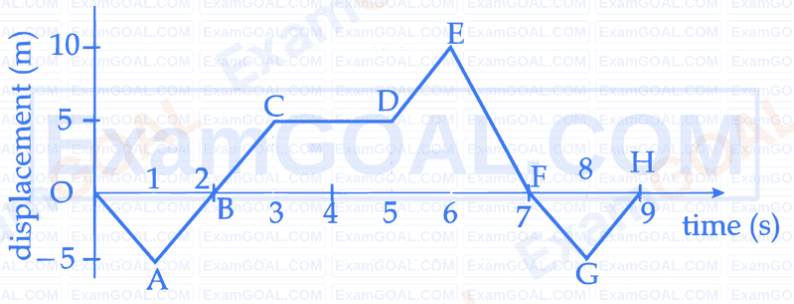
$(\mathrm{g} = 10\ \mathrm{m/s^2})$

(A) 92.5 **(B)** 62.5 **(C)** 20 **(D)** 82.5

Q8. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

The displacement x versus time graph is shown below.



(A) The average velocity during 0 to 3 s is $10\ \mathrm{m/s}$

(B) The average velocity during 3 to 5 s is $0\ \mathrm{m/s}$

(C) The instantaneous velocity at $\mathrm{t} = 2\ \mathrm{s}$ is $5\ \mathrm{m/s}$

(D) The average velocity during 5 to 7 s and instantaneous velocity at $\mathrm{t} = 6.5\ \mathrm{s}$ are equal

(E) The average velocity from $t = 0$ to $t = 9\ \mathrm{s}$ is zero

Choose the correct answer from the options given below :

(A) (B), (C), (D) only **(B)** (B), (D), (E) only

(C) (A), (D), (E) only **(D)** (B), (C), (E) only

Q9. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

A particle moves along the x-axis and has its displacement x varying with time t according to the equation:

$x = \mathrm{c}_0\left(\mathrm{t}^2 - 2\right) + \mathrm{c}(\mathrm{t} - 2)^2$

where c_0 and c are constants of appropriate dimensions.

Then, which of the following statements is correct?

(A) the acceleration of the particle is $2\left(c + c_0\right)$

(B) the acceleration of the particle is $2c_0$

(C) the acceleration of the particle is 2 c

(D) the initial velocity of the particle is $4c$

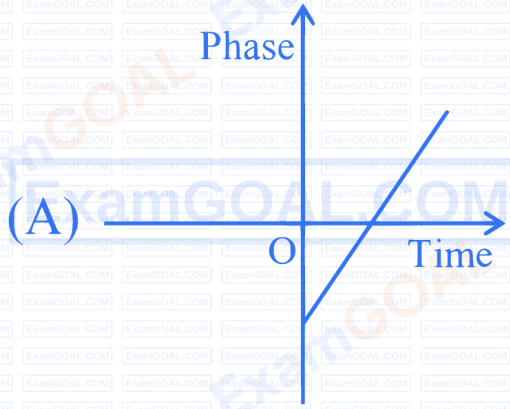
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Q10. JEE Main 2025 (Online) 3rd April Morning Shift

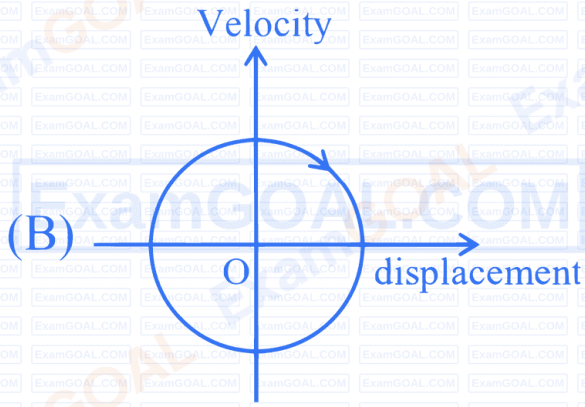
Open solution

Which of the following curves possibly represent one-dimensional motion of a particle?

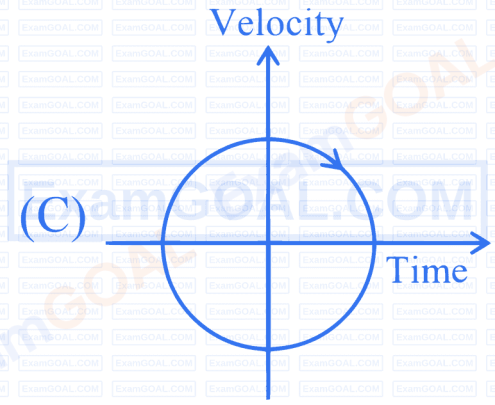
(A)



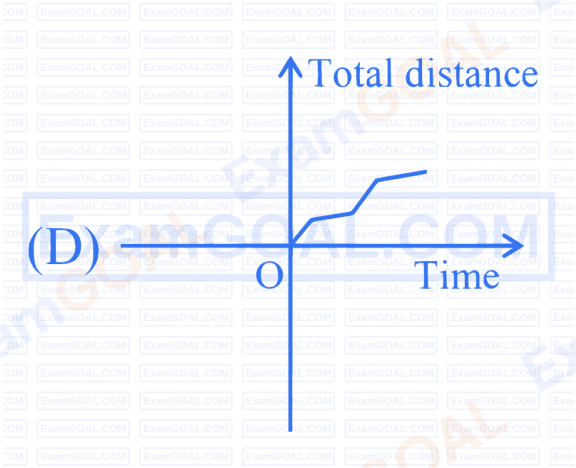
(B)



(C)



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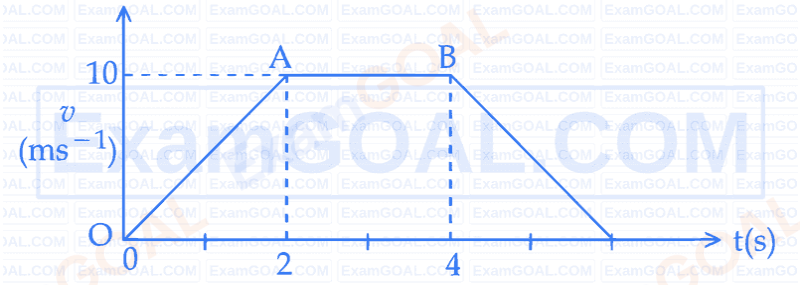
Choose the correct answer from the options given below :

(A) A, B and C only (B) A, C and D only (C) A, B and D only (D) A and B only

Q11. JEE Main 2025 (Online) 28th January Evening Shift

The velocity-time graph of an object moving along a straight line is shown in the figure. What is the distance covered by the object between $t = 0$ to $t = 4s$?

Open solution

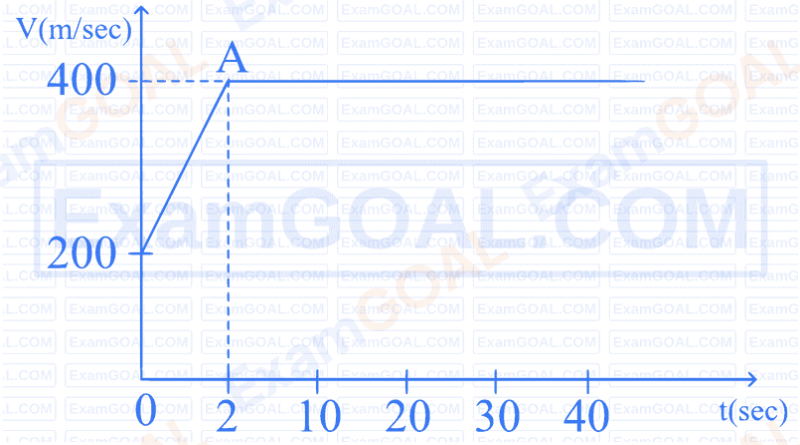


(A) 11 m (B) 30 m (C) 13 m (D) 10 m

Q12. JEE Main 2025 (Online) 23rd January Morning Shift

The motion of an airplane is represented by velocity-time graph as shown below. The distance covered by airplane in the first 30.5 second is_____ km .

Open solution



(A) 12 (B) 6 (C) 3 (D) 9

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Q13. JEE Main 2024 (Online) 9th April Evening Shift

Two cars are travelling towards each other at speed of $20\ \mathrm{m\ s^{-1}}$ each. When the cars are $300\ \mathrm{m}$ apart, both the drivers apply brakes and the cars retard at the rate of $2\ \mathrm{m\ s^{-2}}$. The distance between them when they come to rest is :

Open solution

(A) 25 m (B) 100 m (C) 50 m (D) 200 m

Q14. JEE Main 2024 (Online) 9th April Morning Shift

A particle moving in a straight line covers half the distance with speed $6\ \mathrm{m/s}$. The other half is covered in two equal time intervals with speeds $9\ \mathrm{m/s}$ and $15\ \mathrm{m/s}$ respectively. The average speed of the particle during the motion is :

Open solution

(A) 9.2 m/s (B) 8.8 m/s (C) 10 m/s (D) 8 m/s

Q15. JEE Main 2024 (Online) 6th April Evening Shift

A body projected vertically upwards with a certain speed from the top of a tower reaches the ground in t_1. If it is projected vertically downwards from the same point with the same speed, it reaches the ground in t_2. Time required to reach the ground, if it is dropped from the top of the tower, is :

Open solution

(A) $\sqrt{t_1 + t_2}$ (B) $\sqrt{t_1 t_2}$

(C) $\sqrt{\frac{t_1}{t_2}}$ (D) $\sqrt{t_1 - t_2}$

Q16. JEE Main 2024 (Online) 6th April Morning Shift

A train starting from rest first accelerates uniformly up to a speed of $80\ \mathrm{km/h}$ for time t, then it moves with a constant speed for time $3t$. The average speed of the train for this duration of journey will be (in $\mathrm{km/h}$) :

Open solution

(A) 70 (B) 40 (C) 30 (D) 80

Q17. JEE Main 2024 (Online) 4th April Morning Shift

A body travels $102.5\ \mathrm{m}$ in $\mathrm{n^{th}}$ second and $115.0\ \mathrm{m}$ in $(\mathrm{n}+2)^{\mathrm{th}}$ second. The acceleration is :

Open solution

(A) $6.25\ \mathrm{m/s^2}$ (B) $5\ \mathrm{m/s^2}$ (C) $12.5\ \mathrm{m/s^2}$ (D) $9\ \mathrm{m/s^2}$

Q18. JEE Main 2024 (Online) 1st February Evening Shift

Train A is moving along two parallel rail tracks towards north with speed $72\ \mathrm{km/h}$ and train B is moving towards south with speed $108\ \mathrm{km/h}$. Velocity of train B with respect to A and velocity of ground with respect to B are (in $\mathrm{ms^{-1}}$):

Open solution

(A) -50 and -30 (B) -50 and 30 (C) -30 and 50 (D) 50 and -30

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Q19. JEE Main 2024 (Online) 31st January Morning Shift

The relation between time 't' and distance 'x' is $t = \alpha x^2 + \beta x$, where α and β are constants. The relation between acceleration (a) and velocity (v) is :

(A) $a = -5\alpha v^5$ (B) $a = -3\alpha v^2$ (C) $a = -2\alpha v^3$ (D) $a = -4\alpha v^4$

Open solution

Q20. JEE Main 2024 (Online) 29th January Evening Shift

A particle is moving in a straight line. The variation of position 'x' as a function of time 't' is given as $x = \left(t^3 - 6t^2 + 20t + 15\right)m$. The velocity of the body when its acceleration becomes zero is :

(A) 6 m/s (B) 10 m/s (C) 8 m/s (D) 4 m/s

Open solution

Q21. JEE Main 2024 (Online) 29th January Morning Shift

A body starts moving from rest with constant acceleration covers displacement S_1 in first $(p-1)$ seconds and S_2 in first p seconds. The displacement $\mathrm{S}_1 + \mathrm{S}_2$ will be made in time :

(A) $(2p+1)s$ (B) $(2p-1)s$

(C) $\left(2p^2 - 2p + 1\right)s$ (D) $\sqrt{(2p^2 - 2p + 1)}s$

Open solution

Q22. JEE Main 2023 (Online) 15th April Morning Shift

The position of a particle related to time is given by $x = \left(5t^2 - 4t + 5\right)\mathrm{m}$. The magnitude of velocity of the particle at $t = 2s$ will be :

(A) $14\ \mathrm{ms}^{-1}$ (B) $16\ \mathrm{ms}^{-1}$ (C) $10\ \mathrm{ms}^{-1}$ (D) $06\ \mathrm{ms}^{-1}$

Open solution

Q23. JEE Main 2023 (Online) 13th April Evening Shift

The distance travelled by an object in time t is given by $s = (2.5)t^2$. The instantaneous speed of the object at $\mathrm{t} = 5\ \mathrm{s}$ will be:

(A) $5\ \mathrm{ms}^{-1}$ (B) $12.5\ \mathrm{ms}^{-1}$ (C) $62.5\ \mathrm{ms}^{-1}$ (D) $25\ \mathrm{ms}^{-1}$

Open solution

Q24. JEE Main 2023 (Online) 13th April Evening Shift

A passenger sitting in a train A moving at $90\ \mathrm{km/h}$ observes another train B moving in the opposite direction for $8\ \mathrm{s}$. If the velocity of the train B is $54\ \mathrm{km/h}$, then length of train B is:

(A) 80 m (B) 200 m (C) 120 m (D) 320 m

Open solution

Q25. JEE Main 2023 (Online) 13th April Morning Shift

Two trains 'A' and 'B' of length 'l' and '$4l$' are travelling into a tunnel of length 'L' in parallel tracks from opposite directions with velocities $108\ \mathrm{km/h}$ and $72\ \mathrm{km/h}$, respectively. If train 'A' takes $35\ \mathrm{s}$ less time than train 'B' to cross the tunnel, then length 'L' of tunnel is :

(Given $\mathrm{L} = 60l$)

(A) 900 m (B) 1200 m (C) 1800 m (D) 2700 m

Open solution

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Q26. JEE Main 2023 (Online) 12th April Morning Shift

A ball is thrown vertically upward with an initial velocity of $150\ \mathrm{m/s}$. The ratio of velocity after $3\ \mathrm{s}$ and $5\ \mathrm{s}$ is $\frac{x+1}{x}$. The value of x is _____.

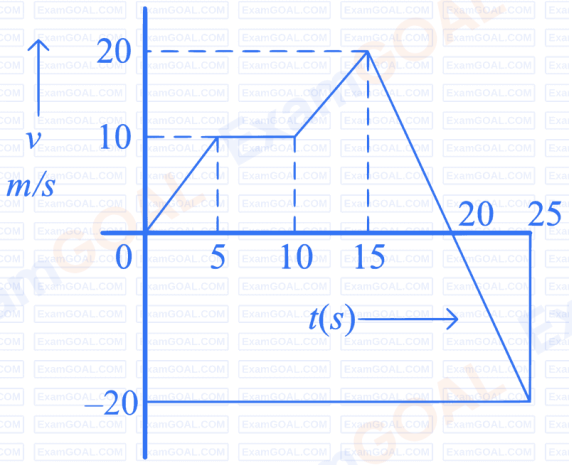
{ take, $g = 10\ \mathrm{m/s^2}$}

Open solution

(A) -5 (B) 10 (C) 5 (D) 6

Q27. JEE Main 2023 (Online) 11th April Morning Shift

From the $\mathrm{v} - t$ graph shown, the ratio of distance to displacement in $25\ \mathrm{s}$ of motion is:



Open solution

(A) 1 (B) $\frac{3}{5}$ (C) $\frac{1}{2}$ (D) $\frac{5}{3}$

Q28. JEE Main 2023 (Online) 10th April Evening Shift

A person travels x distance with velocity v_1 and then x distance with velocity v_2 in the same direction. The average velocity of the person is v, then the relation between v, v_1 and v_2 will be.

Open solution

(A) $\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2$ (B) $V = \frac{v_1 + V_2}{2}$

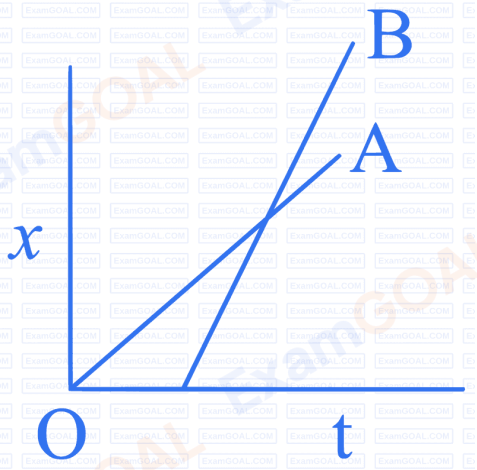
(C) $\frac{1}{\mathrm{v}} = \frac{1}{\mathrm{v}_1} + \frac{1}{\mathrm{v}_2}$ (D) $\frac{2}{\mathrm{V}} = \frac{1}{\mathrm{v}_1} + \frac{1}{\mathrm{v}_2}$

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Q29. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

The position-time graphs for two students A and B returning from the school to their homes are shown in figure.



(A) A lives closer to the school

(B) B lives closer to the school

(C) A takes lesser time to reach home

(D) A travels faster than B

(E) B travels faster than A

Choose the correct answer from the options given below :

(A) (A), (C) and (D) only

(B) (A) and (E) only

(C) (A), (C) and (E) only

(D) (B) and (E) only

Q30. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

Given below are two statements

Statement I : Area under velocity- time graph gives the distance travelled by the body in a given time.

Statement II : Area under acceleration- time graph is equal to the change in velocity in the given time.

In the light of given statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are False.

(B) Both Statement I and Statement II are true.

(C) Statement I is incorrect but Statement II is true.

(D) Statement I is correct but Statement II is false.

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Q31. JEE Main 2023 (Online) 6th April Evening Shift

A particle starts with an initial velocity of $10.0\ \mathrm{ms}^{-1}$ along x-direction and accelerates uniformly at the rate of $2.0\ \mathrm{ms}^{-2}$. The time taken by the particle to reach the velocity of $60.0\ \mathrm{ms}^{-1}$ is _____.

(A) 30s (B) 6s (C) 3s (D) 25s

Open solution

Q32. JEE Main 2023 (Online) 1st February Morning Shift

An object moves with speed v_1, v_2 and v_3 along a line segment AB, BC and CD respectively as shown in figure. Where AB = BC and AD = 3AB, then average speed of the object will be:



(A) $\frac{v_1v_2v_3}{3(v_1v_2+v_2v_3+v_3v_1)}$ (B) $\frac{(v_1+v_2+v_3)}{3}$

(C) $\frac{(v_1+v_2+v_3)}{3v_1v_2v_3}$ (D) $\frac{3v_1v_2v_3}{(v_1v_2+v_2v_3+v_3v_1)}$

Open solution

Q33. JEE Main 2023 (Online) 30th January Evening Shift

A vehicle travels $4\ \mathrm{km}$ with speed of $3\ \mathrm{km/h}$ and another $4\ \mathrm{km}$ with speed of $5\ \mathrm{km/h}$, then its average speed is

(A) $3.75\ \mathrm{km/h}$ (B) $4.25\ \mathrm{km/h}$ (C) $3.50\ \mathrm{km/h}$ (D) $4.00\ \mathrm{km/h}$

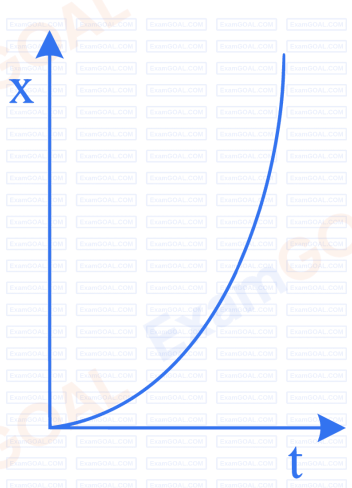
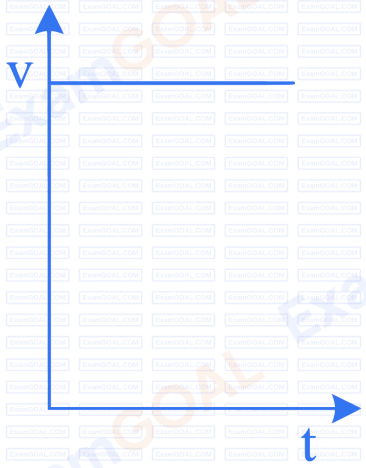
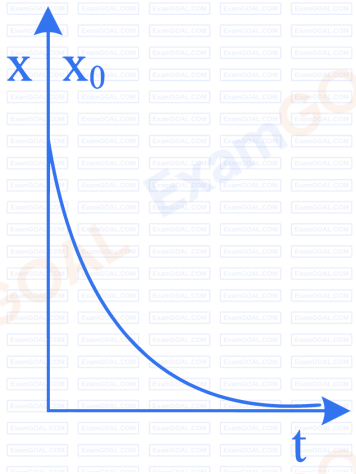
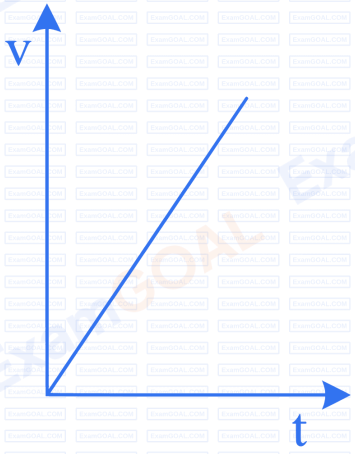
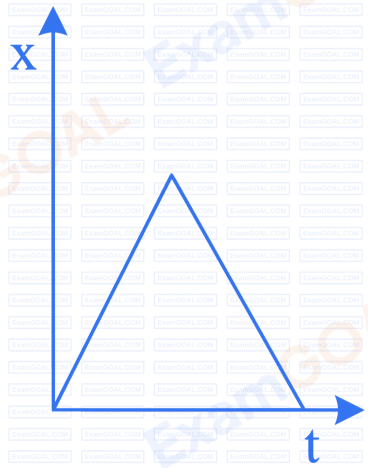
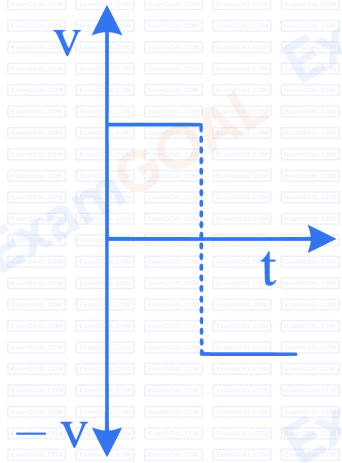
Open solution

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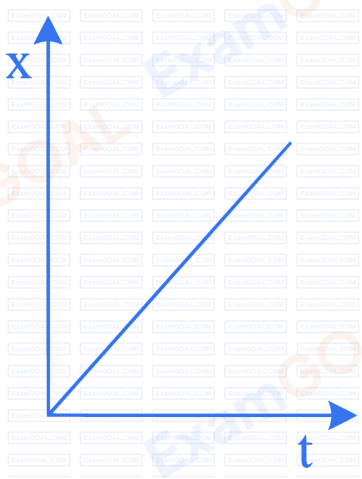
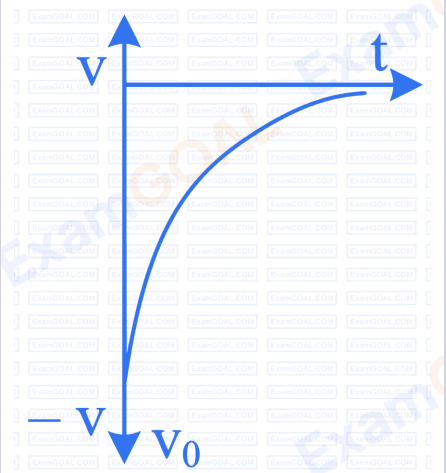
Q34. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

Match Column-I with Column-II :

	Column-I ($_{\mathfrak{X}}$-t graphs)		Column-II ($_{\Omega}$-t graphs)
A.		I.	
B.		II.	
C.		III.	

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	Column-I ($_{\mathfrak{X}}$-t graphs)		Column-II ($_{\Omega}$-t graphs)
D.		IV.	

Choose the correct answer from the options given below:

(A) A- II, B-III, C-IV, D-I

(B) A- II, B-IV, C-III, D-I

(C) A- I, B-III, C-IV, D-II

(D) A- I, B-II, C-III, D-IV

Q35. JEE Main 2023 (Online) 25th January Evening Shift

The distance travelled by a particle is related to time t as $x = 4t^2$. The velocity of the particle at t=5s is :-

(A) $25\ ms^{-1}$

(B) $20\ ms^{-1}$

(C) $8\ ms^{-1}$

(D) $40\ ms^{-1}$

Open solution

Q36. JEE Main 2023 (Online) 25th January Morning Shift

A car travels a distance of 'x' with speed v_1 and then same distance 'x' with speed v_2 in the same direction. The average speed of the car is :

(A) $\frac{v_1 v_2}{2(v_1+v_2)}$

(B) $\frac{2v_1 v_2}{v_1+v_2}$

(C) $\frac{2x}{v_1+v_2}$

(D) $\frac{v_1+v_2}{2}$

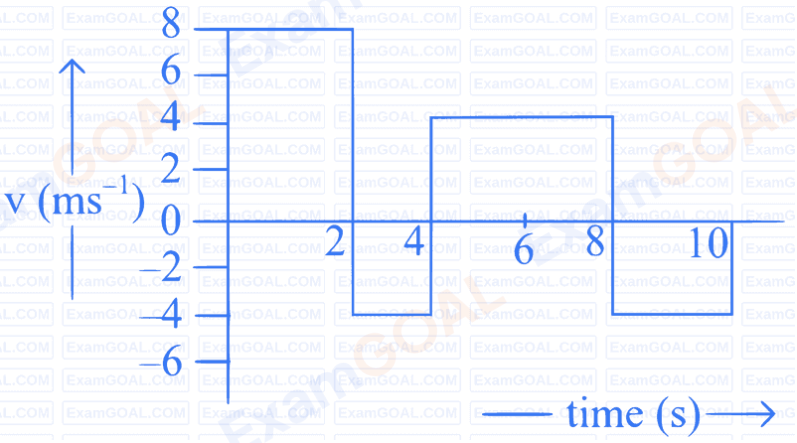
Open solution

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Q37. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

The velocity time graph of a body moving in a straight line is shown in the figure.



The ratio of displacement to distance travelled by the body in time 0 to 10s is :

(A) 1 : 1 (B) 1 : 3 (C) 1 : 4 (D) 1 : 2

Q38. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

Two trucks A and B are approaching each other on a straight path with the velocities of $16\ \mathrm{m/s}$ and $20\ \mathrm{m/s}$ respectively. When they are $200\ \mathrm{m}$ apart their drivers see each other and applies the breaks simultaneously. If the truck A decelerates with $2\ \mathrm{m/s^2}$ and truck B decelerates with $4\ \mathrm{m/s^2}$, what is the distance between them when they finally stop?

(A) $14\ \mathrm{m}$ (B) $114\ \mathrm{m}$ (C) $86\ \mathrm{m}$ (D) $64\ \mathrm{m}$

Q39. JEE Main 2022 (Online) 29th July Evening Shift

Open solution

A juggler throws balls vertically upwards with same initial velocity in air. When the first ball reaches its highest position, he throws the next ball. Assuming the juggler throws n balls per second, the maximum height the balls can reach is

(A) g/2n (B) g/n (C) 2gn (D) $g/2n^2$

Q40. JEE Main 2022 (Online) 29th July Evening Shift

Open solution

A ball is released from a height h. If t_1 and t_2 be the time required to complete first half and second half of the distance respectively. Then, choose the correct relation between t_1 and t_2.

(A) $t_1 = (\sqrt{2})t_2$ (B) $t_1 = (\sqrt{2}-1)t_2$ (C) $t_2 = (\sqrt{2}+1)t_1$ (D) $t_2 = (\sqrt{2}-1)t_1$

Q41. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

A ball is thrown up vertically with a certain velocity so that, it reaches a maximum height h. Find the ratio of the times in which it is at height $\frac{h}{3}$ while going up and coming down respectively.

(A) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$ (B) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

(C) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (D) $\frac{1}{3}$

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Q42. JEE Main 2022 (Online) 29th July Morning Shift

If $t = \sqrt{x} + 4$, then $\left(\frac{dx}{dt}\right)_{t=4}$ is :

(A) 4 (B) zero (C) 8 (D) 16

Open solution

Q43. JEE Main 2022 (Online) 28th July Morning Shift

An NCC parade is going at a uniform speed of $9\ \mathrm{km/h}$ under a mango tree on which a monkey is sitting at a height of $19.6\ \mathrm{m}$. At any particular instant, the monkey drops a mango. A cadet will receive the mango whose distance from the tree at time of drop is: (Given $g = 9.8\ \mathrm{m/s^2}$)

(A) 5 m (B) 10 m (C) 19.8 m (D) 24.5 m

Open solution

Q44. JEE Main 2022 (Online) 27th July Evening Shift

The velocity of the bullet becomes one third after it penetrates 4 cm in a wooden block. Assuming that bullet is facing a constant resistance during its motion in the block. The bullet stops completely after travelling at (4 + x) cm inside the block. The value of x is :

(A) 2.0 (B) 1.0 (C) 0.5 (D) 1.5

Open solution

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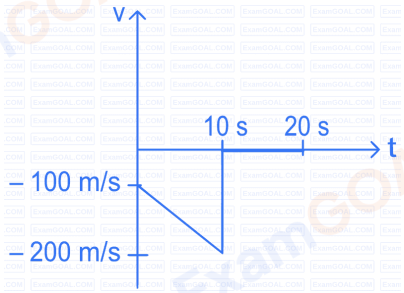
Q45. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

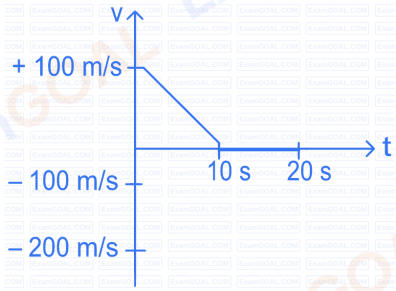
A bullet is shot vertically downwards with an initial velocity of $100\ \mathrm{m/s}$ from a certain height. Within 10 s, the bullet reaches the ground and instantaneously comes to rest due to the perfectly inelastic collision. The velocity-time curve for total time $\mathrm{t} = 20\ \mathrm{s}$ will be:

(Take g = 10 m/s^2).

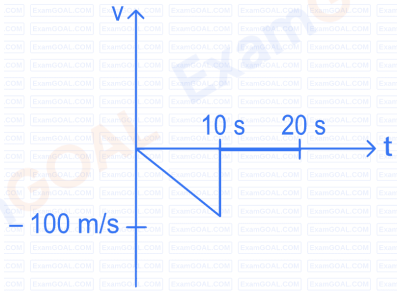
(A)



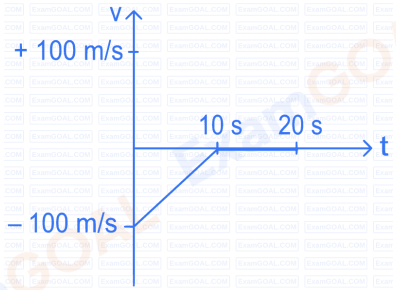
(B)



(C)



(D)



Q46. JEE Main 2022 (Online) 29th June Evening Shift

Open solution

A small toy starts moving from the position of rest under a constant acceleration. If it travels a distance of 10m in t s, the distance travelled by the toy in the next t s will be :

(A) 10 m (B) 20 m (C) 30 m (D) 40 m

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Q47. JEE Main 2022 (Online) 29th June Morning Shift

Two balls A and B are placed at the top of 180 m tall tower. Ball A is released from the top at t = 0 s. Ball B is thrown vertically down with an initial velocity 'u' at t = 2 s. After a certain time, both balls meet 100 m above the ground. Find the value of 'u' in ms^{-1}. [use g = 10 ms^{-2}] :

Open solution

(A) 10 (B) 15 (C) 20 (D) 30

Q48. JEE Main 2022 (Online) 25th June Evening Shift

Two buses P and Q start from a point at the same time and move in a straight line and their positions are represented by $X_P(t) = \alpha t + \beta t^2$ and $X_Q(t) = ft - t^2$. At what time, both the buses have same velocity?

Open solution

(A) $\frac{\alpha - f}{1+\beta}$ (B) $\frac{\alpha + f}{2(\beta - 1)}$

(C) $\frac{\alpha + f}{2(1+\beta)}$ (D) $\frac{f - \alpha}{2(1+\beta)}$

Q49. JEE Main 2021 (Online) 27th August Evening Shift

Water drops are falling from a nozzle of a shower onto the floor, from a height of 9.8 m. The drops fall at a regular interval of time. When the first drop strikes the floor, at that instant, the third drop begins to fall. Locate the position of second drop from the floor when the first drop strikes the floor.

Open solution

(A) 4.18 m (B) 2.94 m (C) 2.45 m (D) 7.35 m

Q50. JEE Main 2021 (Online) 27th July Morning Shift

A ball is thrown up with a certain velocity so that it reaches a height 'h'. Find the ratio of the two different times of the ball reaching $\frac{h}{3}$ in both the directions.

Open solution

(A) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$ (B) $\frac{1}{3}$

(C) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$ (D) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

Q51. JEE Main 2021 (Online) 25th July Evening Shift

The relation between time t and distance x for a moving body is given as $t = mx^2 + nx$, where m and n are constants. The retardation of the motion is : (When v stands for velocity)

Open solution

(A) $2mv^3$ (B) $2mnv^3$ (C) $2nv^3$ (D) $2n^2v^3$

Q52. JEE Main 2021 (Online) 25th July Evening Shift

A balloon was moving upwards with a uniform velocity of 10 m/s. An object of finite mass is dropped from the balloon when it was at a height of 75 m from the ground level. The height of the balloon from the ground when object strikes the ground was around :

Open solution

(take the value of g as 10 m/s^2)

(A) 300 m (B) 200 m (C) 125 m (D) 250 m

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Q53. JEE Main 2021 (Online) 25th July Evening Shift

The instantaneous velocity of a particle moving in a straight line is given as $V = \alpha t + \beta t^2$, where α and β are constants. The distance travelled by the particle between 1s and 2s is :

Open solution

(A) $3\alpha + 7\beta$

(B) $\frac{3}{2}\alpha + \frac{7}{3}\beta$

(C) $\frac{\alpha}{2} + \frac{\beta}{3}$

(D) $\frac{3}{2}\alpha + \frac{7}{2}\beta$

Q54. JEE Main 2021 (Online) 25th July Morning Shift

Water droplets are coming from an open tap at a particular rate. The spacing between a droplet observed at 4th second after its fall to the next droplet is 34.3 m. At what rate the droplets are coming from the tap ? (Take g = 9.8 m/s^2)

Open solution

(A) 3 drops / 2 seconds

(B) 2 drops / second

(C) 1 drop / second

(D) 1 drop / 7 seconds

Q55. JEE Main 2021 (Online) 20th July Evening Shift

A boy reaches the airport and finds that the escalator is not working. He walks up the stationary escalator in time t_1. If he remains stationary on a moving escalator then the escalator takes him up in time t_2. The time taken by him to walk up on the moving escalator will be :

Open solution

(A) $\frac{t_1 t_2}{t_2 - t_1}$

(B) $\frac{t_1 + t_2}{2}$

(C) $\frac{t_1 t_2}{t_2 + t_1}$

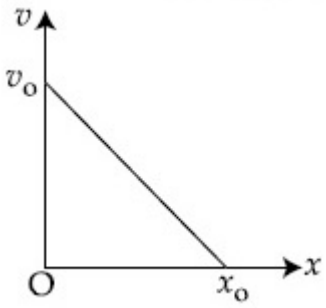
(D) $t_2 - t_1$

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Q56. JEE Main 2021 (Online) 18th March Evening Shift

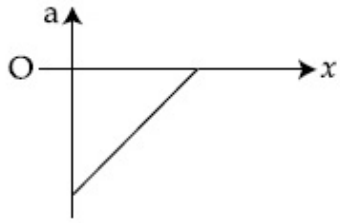
Open solution

The velocity – displacement graph of a particle is shown in the figure.

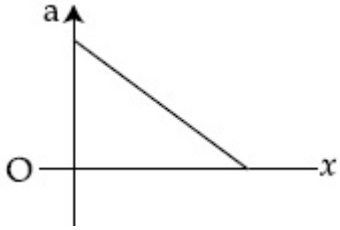


The acceleration – displacement graph of the same particle is represented by :

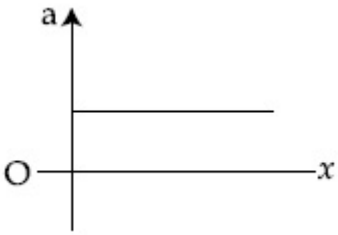
(A)



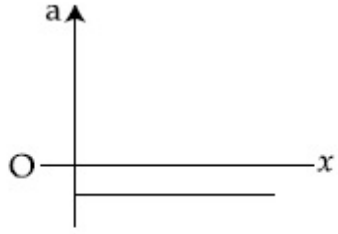
(B)



(C)



(D)



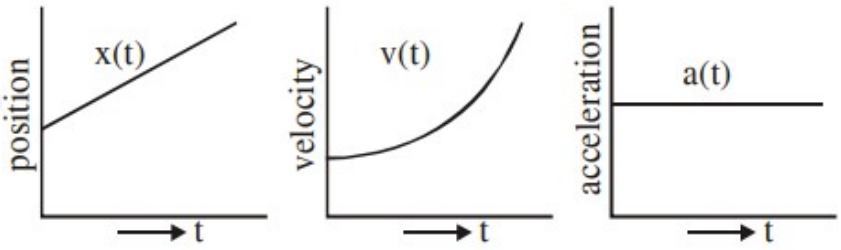
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Q57. JEE Main 2021 (Online) 18th March Morning Shift

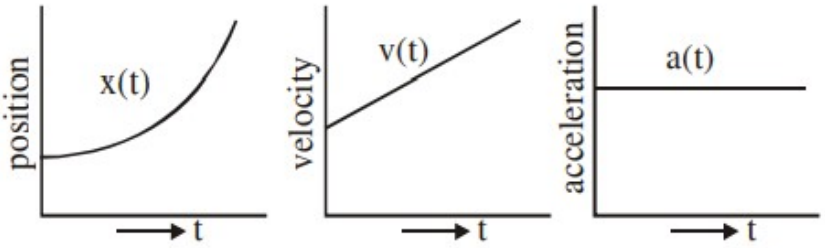
The position, velocity and acceleration of a particle moving with a constant acceleration can be represented by :

Open solution

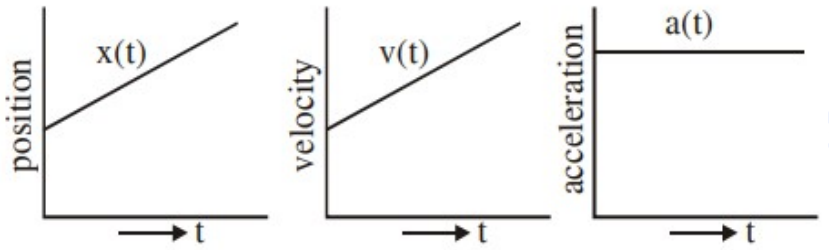
(A)



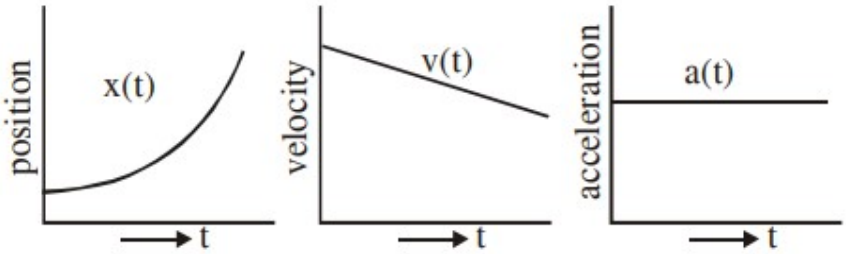
(B)



(C)



(D)



Q58. JEE Main 2021 (Online) 17th March Evening Shift

The velocity of a particle is $v = v_0 + gt + Ft^2$. Its position is x = 0 at t = 0; then its displacement after time (t = 1) is :

Open solution

(A) $v_0 + g + f$

(B) $v_0 + \frac{g}{2} + \frac{F}{3}$

(C) $v_0 + 2g + 3F$

(D) $v_0 + \frac{g}{2} + F$

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Q59. JEE Main 2021 (Online) 17th March Evening Shift

A rubber ball is released from a height of 5 m above the floor. It bounces back repeatedly, always rising to $\frac{81}{100}$ of the height through which it falls. Find the average speed of the ball. (Take g = 10 ms $^{-2}$)

Open solution

(A) 2.50 ms^{-1} (B) 3.0 ms^{-1} (C) 2.0 ms^{-1} (D) 3.50 ms^{-1}

Q60. JEE Main 2021 (Online) 17th March Morning Shift

A car accelerates from rest at a constant rate α for some time after which it decelerates at a constant rate β to come to rest. If the total time elapsed is t seconds, the total distance travelled is :

Open solution

(A) $\frac{4\alpha\beta}{(\alpha+\beta)}t^2$ (B) $\frac{2\alpha\beta}{(\alpha+\beta)}t^2$

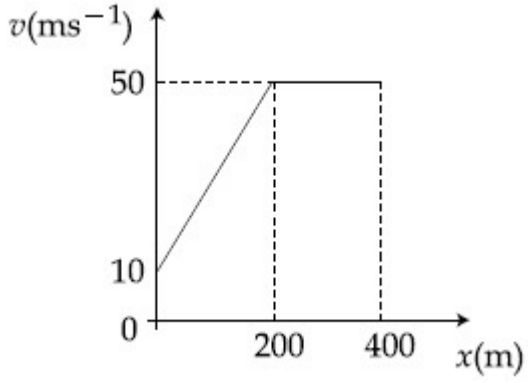
(C) $\frac{\alpha\beta}{2(\alpha+\beta)}t^2$ (D) $\frac{\alpha\beta}{4(\alpha+\beta)}t^2$

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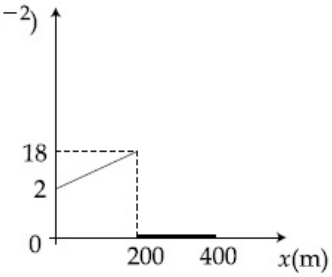
Q61. JEE Main 2021 (Online) 16th March Morning Shift

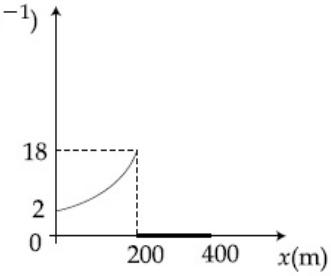
Open solution

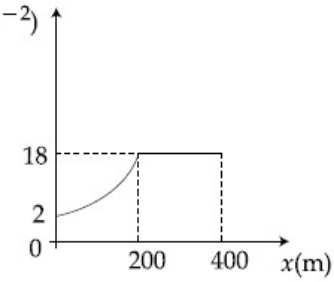
The velocity-displacement graph describing the motion of bicycle is shown in the figure.



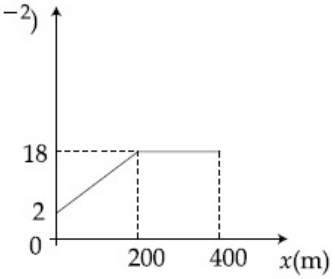
The acceleration-displacement graph of the bicycle's motion is best described by :

(A)


(B)


(C)


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(D) 

Q62. JEE Main 2021 (Online) 26th February Evening Shift

A scooter accelerates from rest for time t_1 at constant rate a_1 and then retards at constant rate a_2 for time t_2 and comes to rest. The correct value of $\frac{t_1}{t_2}$ will be :

Open solution

(A) $\frac{a_1+a_2}{a_2}$ (B) $\frac{a_1+a_2}{a_1}$ (C) $\frac{a_2}{a_1}$ (D) $\frac{a_1}{a_2}$

Q63. JEE Main 2021 (Online) 25th February Evening Shift

A stone is dropped from the top of a building. When it crosses a point 5 m below the top, another stone starts to fall from a point 25 m below the top. Both stones reach the bottom of building simultaneously. The height of the building is :

Open solution

(A) 50 m (B) 25 m (C) 45 m (D) 35 m

Q64. JEE Main 2021 (Online) 25th February Morning Shift

An engine of a train, moving with uniform acceleration, passes the signal-post with velocity u and the last compartment with velocity v. The velocity with which middle point of the train passes the signal post is :

Open solution

(A) $\frac{u+v}{2}$ (B) $\sqrt{\frac{v^2-u^2}{2}}$

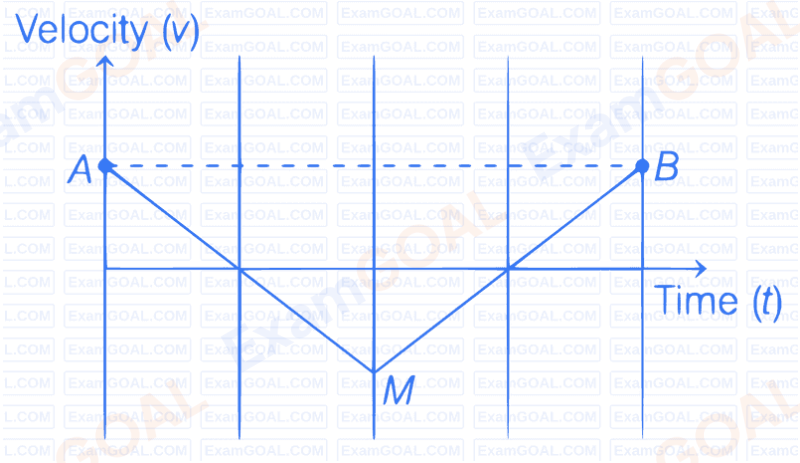
(C) $\frac{v-u}{2}$ (D) $\sqrt{\frac{v^2+u^2}{2}}$

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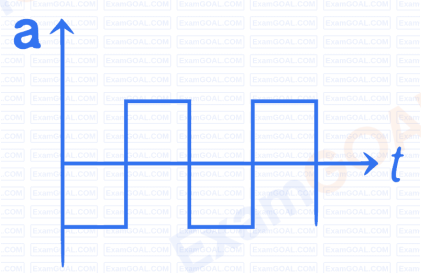
Q65. JEE Main 2021 (Online) 24th February Morning Shift

If the velocity-time graph has the shape AMB, what would be the shape of the corresponding acceleration-time graph?

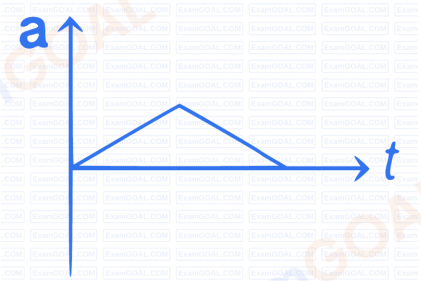
Open solution



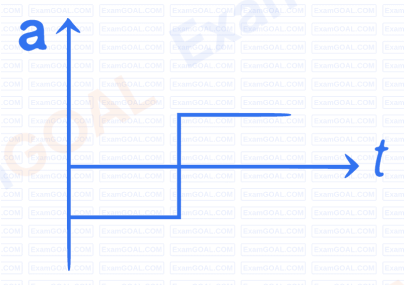
(A)



(B)

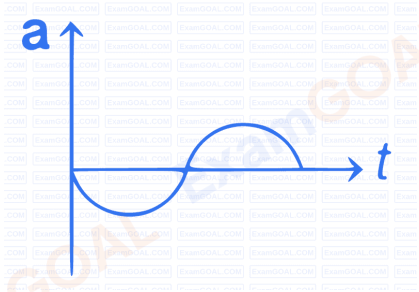


(C)



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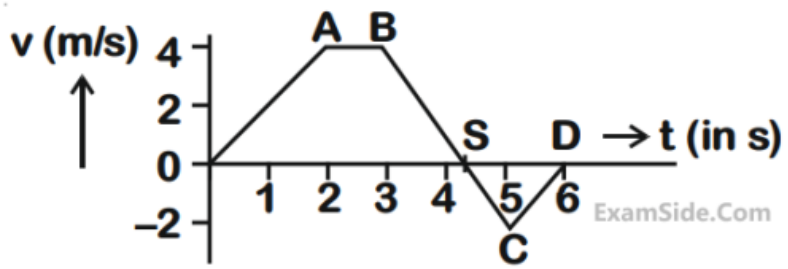
(D)



Q66. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

The velocity (v) and time (t) graph of a body in a straight line motion is shown in the figure. The point S is at 4.333 seconds. The total distance covered by the body in 6 s is :



(A) 12 m (B) 11 m (C) $\frac{49}{4}$ m (D) $\frac{37}{3}$ m

Q67. JEE Main 2020 (Online) 5th September Morning Slot

Open solution

A helicopter rises from rest on the ground vertically upwards with a constant acceleration g. A food packet is dropped from the helicopter when it is at a height h. The time taken by the packet to reach the ground is close to :
[g is the acceleration due to gravity]

(A) $t = 3.4\sqrt{\left(\frac{h}{g}\right)}$ (B) $t = 1.8\sqrt{\left(\frac{h}{g}\right)}$

(C) $t = \sqrt{\frac{2h}{3g}}$ (D) $t = \frac{2}{3}\sqrt{\left(\frac{h}{g}\right)}$

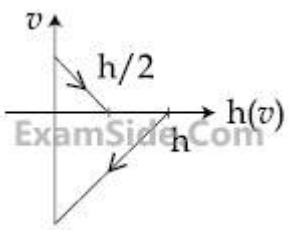
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Q68. JEE Main 2020 (Online) 4th September Morning Slot

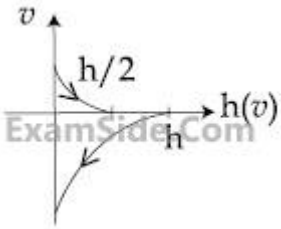
Open solution

A Tennis ball is released from a height h and after freely falling on a wooden floor it rebounds and reaches height $\frac{h}{2}$. The velocity versus height of the ball during its motion may be represented graphically by :
(graph are drawn schematically and on not to scale)

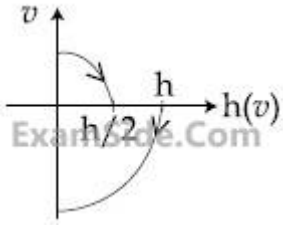
(A)



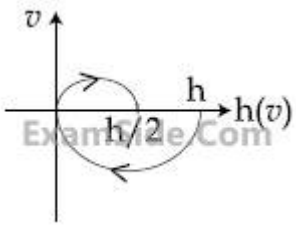
(B)



(C)



(D)



Q69. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution

Train A and train B are running on parallel tracks in the opposite directions with speeds of 36 km/hour and 72 km/hour, respectively. A person is walking in train A in the direction opposite to its motion with a speed of 1.8 km/ hour. Speed (in ms^{-1}) of this person as observed from train B will be close to :
(take the distance between the tracks as negligible)

(A) 30.5 ms^{-1} (B) 29.5 ms^{-1} (C) 31.5 ms^{-1} (D) 28.5 ms^{-1}

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Q70. JEE Main 2019 (Online) 12th April Evening Slot

A particle is moving with speed v = b$\sqrt{x}$ along positive x-axis. Calculate the speed of the particle at time t = τ(assume that the particle is at origin t = 0)

(A) $\frac{b^2\tau}{\sqrt{2}}$ (B) $b^2\tau$ (C) $\frac{b^2\tau}{2}$ (D) $\frac{b^2\tau}{4}$

Open solution

Q71. JEE Main 2019 (Online) 9th April Evening Slot

The position of a particle as a function of time t, is given by

$x(t) = at + bt^2 - ct^3$

where a, b and c are constants. When the particle attains zero acceleration, then its velocity will be :

(A) $a + \frac{b^2}{c}$ (B) $a + \frac{b^2}{4c}$ (C) $a + \frac{b^2}{3c}$ (D) $a + \frac{b^2}{2c}$

Open solution

Q72. JEE Main 2019 (Online) 9th April Evening Slot

The position vector of a particle changes with time according to the relation

$\overrightarrow{r}(t) = 15t^2\hat{i} + (4 - 20t^2)\hat{j}$

What is the magnitude of the acceleration at t = 1 ?

(A) 50 (B) 25 (C) 40 (D) 100

Open solution

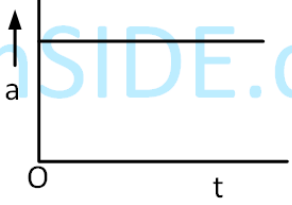
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Q73. JEE Main 2019 (Online) 8th April Evening Slot

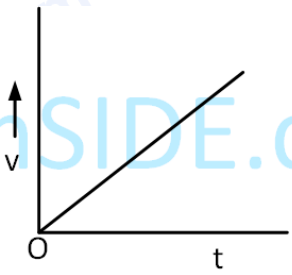
Open solution

A particle starts from origin O from rest and moves with a uniform acceleration along the positive x-axis. Identify all figures that correctly represent the motion qualitatively. (a = acceleration, v = velocity, x = displacement, t = time)

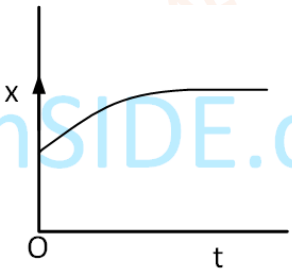
(A)



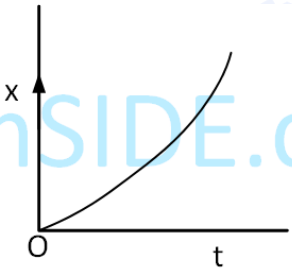
(B)



(C)



(D)



(A) (B), (C) **(B)** (A) **(C)** (A), (B), (C) **(D)** (A), (B), (D)

Q74. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

A passenger train of length 60 m travels at a speed of 80 km/hr. Another freight train of length 120 m travels at a speed of 30 km/hr. The ratio of times taken by the passenger train to completely cross the freight train when : (i) they are moving in the same direction , and (ii) in the opposite direction is :

(A) $\frac{25}{11}$ **(B)** $\frac{3}{2}$ **(C)** $\frac{11}{5}$ **(D)** $\frac{5}{2}$

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Q75. JEE Main 2019 (Online) 11th January Evening Slot

A particle moves from the point $\left(2.0\hat{i}+4.0\hat{j}\right)$ m, at t = 0, with an initial velocity $\left(5.0\hat{i}+4.0\hat{j}\right)$ ms^{-1}. It is acted upon by a constant force which produces a constant acceleration $\left(4.0\hat{i}+4.0\hat{j}\right)$ ms^{-2}. What is the distance of the particle from the origin at time 2 s?

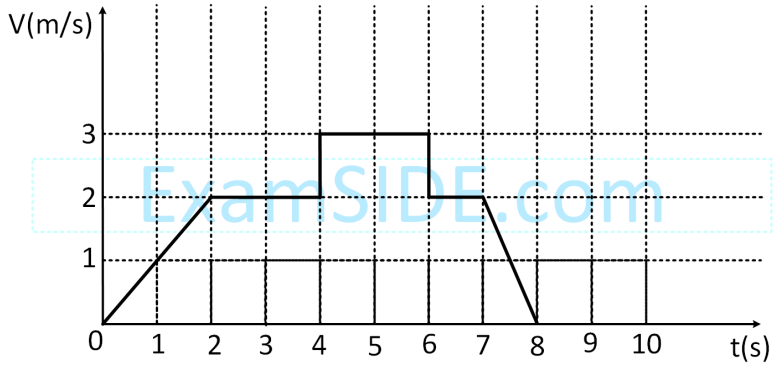
Open solution

(A) 15 m (B) $20\sqrt{2}$ m (C) $10\sqrt{2}$ m (D) 5 m

Q76. JEE Main 2019 (Online) 10th January Evening Slot

A particle starts from the origin at time t = 0 and moves along the positive x-axis. The graph of velocity with respect to time is shown in figure. What is the position of the particle at time t = 5s ?

Open solution



(A) 3 m (B) 9 m (C) 10 m (D) 6 m

Q77. JEE Main 2019 (Online) 9th January Evening Slot

In a car race on straight road, car A takes a time t less than car B at the finish and passes finishing point with a speed 'v' more than that of car B. Both the cars start from rest and travel with constant acceleration a_1 and a_2 respectively. Then 'v' is equal to :

Open solution

(A) $\frac{2a_1a_2}{a_1+a_2}t$ (B) $\sqrt{2a_1a_2}t$
(C) $\sqrt{a_1a_2}t$ (D) $\frac{a_1+a_2}{2}t$

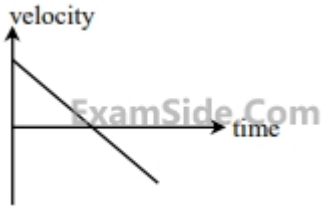
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Q78. JEE Main 2018 (Offline)

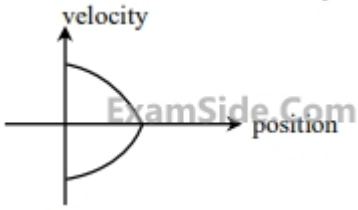
All the graphs below are intended to represent the same motion. One of them does it incorrectly. Pick it up.

Open solution

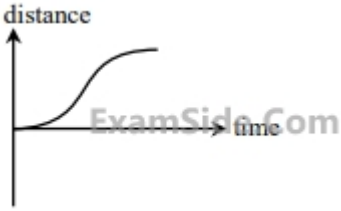
(A)



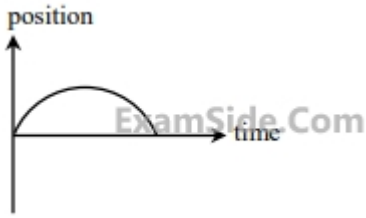
(B)



(C)



(D)



Q79. JEE Main 2018 (Online) 15th April Morning Slot

An automobile, travelling at 40 km/h, can be stopped at a distance of 40 m by applying brakes. If the same automobile is travelling at 80 km/h, the minimum stopping distance, in metres, is (assume no skidding) :

Open solution

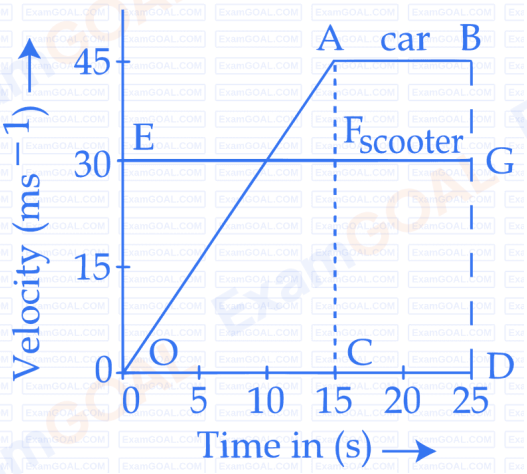
(A) 45 m (B) 100 m (C) 150 m (D) 160 m

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Q80. JEE Main 2018 (Online) 15th April Morning Slot

The velocity-time graphs of a car and a scooter are shown in the figure. (i) The difference between the distance travelled by the car and the scooter in $15\ s$ and (ii) the time at which the car will catch up with the scooter are, respectively.

Open solution



(A) $112.5\ m$ and $22.5\ s$

(B) $337.5\ m$ and $25\ s$

(C) $112.5\ m$ and $15\ s$

(D) $225.5\ m$ and $10\ s$

Q81. JEE Main 2017 (Online) 9th April Morning Slot

A car is standing 200 m behind a bus, which is also at rest. The two start moving at the same instant but with different forward accelerations. The bus has acceleration 2 m/s^2 and the car has acceleration 4 m/s^2 . The car will catch up with the bus after a time of :

Open solution

(A) $\sqrt{110}\ s$

(B) $\sqrt{120}\ s$

(C) $10\sqrt{2}\ s$

(D) 15 s

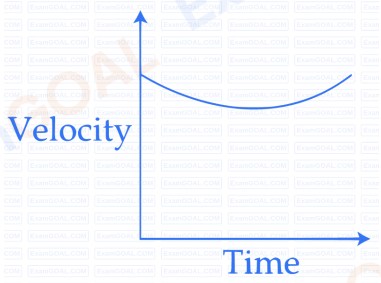
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Q82. JEE Main 2017 (Online) 8th April Morning Slot

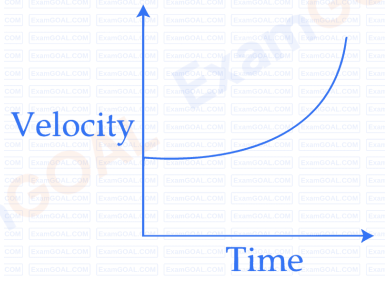
Which graph corresponds to an object moving with a constant negative acceleration and a positive velocity ?

Open solution

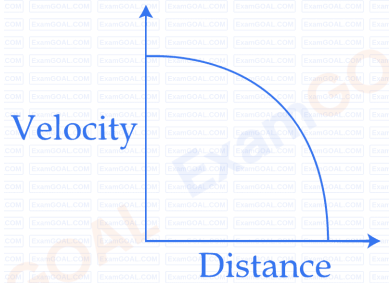
(A)



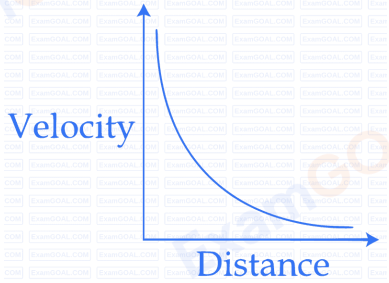
(B)



(C)



(D)



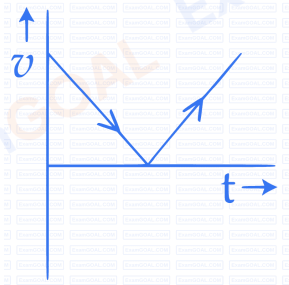
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Q83. JEE Main 2017 (Offline)

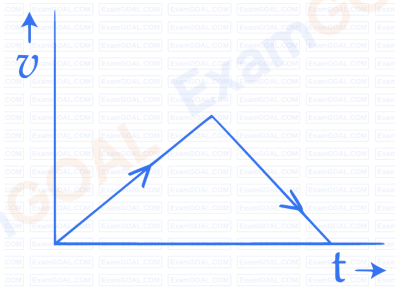
A body is thrown vertically upwards. Which one of the following graphs correctly represent the velocity vs time?

Open solution

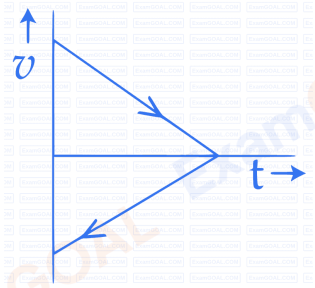
(A)



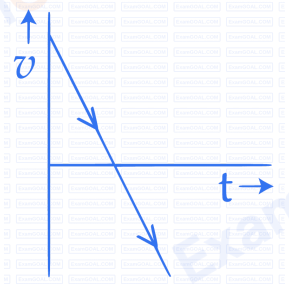
(B)



(C)



(D)



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Q84. JEE Main 2015 (Offline)

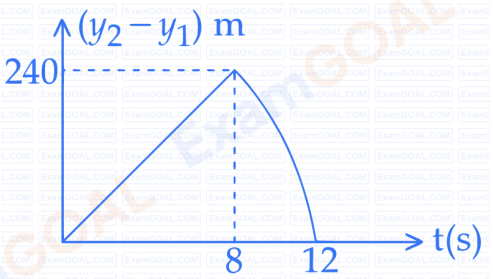
Open solution

Two stones are thrown up simultaneously from the edge of a cliff $240\ m$ high with initial speed of $10\ m/s$ and $40\ m/s$ respectively. Which of the following graph best represents the time variation of relative position of the second stone with respect to the first ?

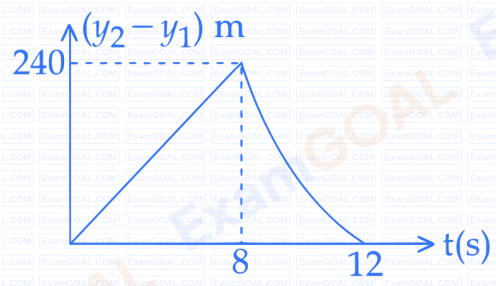
(Assume stones do not rebound after hitting the ground and neglect air resistance, take $g = 10m/s^2$)

(The figures are schematic and not drawn to scale)

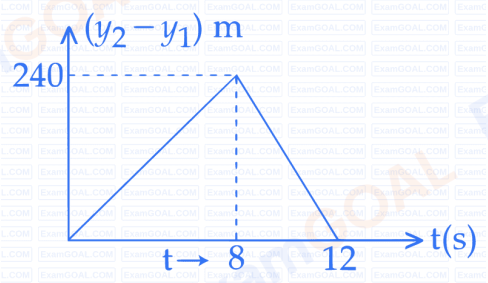
(A)



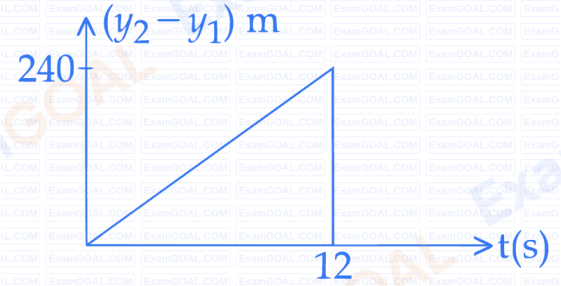
(B)



(C)



(D)



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Q85. JEE Main 2014 (Online) 12th April Morning Slot

A Person climbs up a stalled escalator in 60 s. If standing on the same but escalator running with constant velocity he takes 40 s. How much time is taken by the person to walk up the moving escalator ?

Open solution

(A) 37 s (B) 27 s (C) 24 s (D) 45 s

Q86. JEE Main 2014 (Offline)

From a tower of height H, a particle is thrown vertically upwards with a speed u. The time taken by the particle, to hit the ground, is n times that taken by it to reach the highest point of its path. The relation between H, u and n is:

Open solution

(A) $2gH = n^2u^2$ (B) $gH = (n - 2)^2u^2$ (C) $2gH = nu^2(n - 2)$ (D) $gH = (n - 2)u^2$

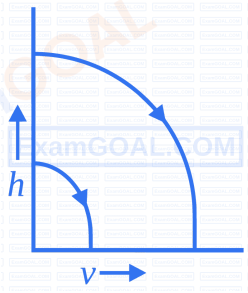
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Q87. AIEEE 2012 (Online) 26th May

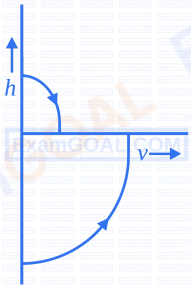
A ball is dropped vertically downwards from a height h above the ground. It hits the ground inelastically and bounces up vertically. Neglecting subsequent motion and air resistance, which of the following graph represents variation between speed (v) and height (h) correctly?

Open solution

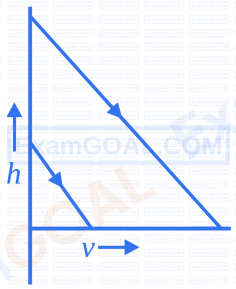
(A)



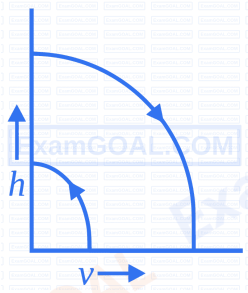
(B)



(C)



(D)



Q88. AIEEE 2012 (Online) 19th May

A goods train accelerating uniformly on a straight railway track, approaches an electric pole standing on the side of track. Its engine passes the pole with velocity u and the guard's room passes with velocity v. The middle wagon of the train passes the pole with a velocity.

Open solution

(A) $\frac{u+v}{2}$

(B) $\frac{1}{2}\sqrt{u^2+v^2}$

(C) $\sqrt{uv}$

(D) $\sqrt{\left(\frac{u^2+v^2}{2}\right)}$

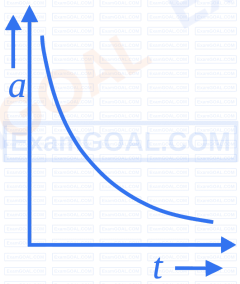
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Q89. AIEEE 2012 (Online) 12th May

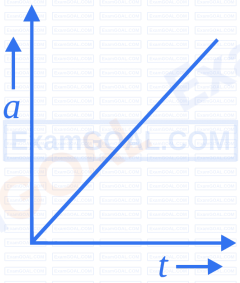
The distance travelled by a body moving along a line in time t is proportional to t^3. The acceleration-time (a, t) graph for the motion of the body will be

Open solution

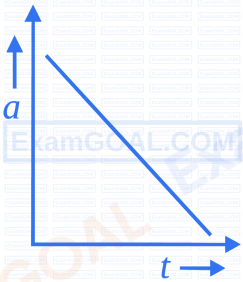
(A)



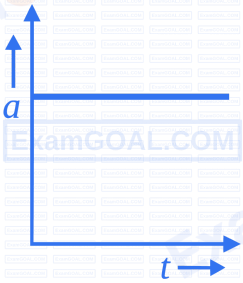
(B)



(C)



(D)

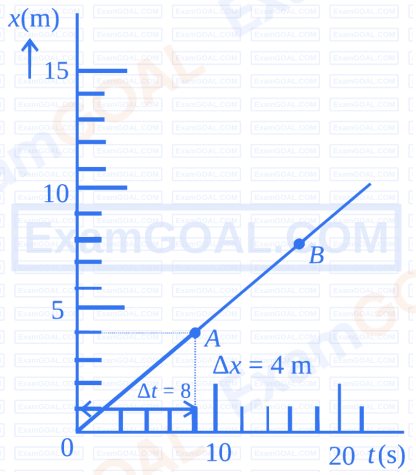


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Q90. AIEEE 2012 (Online) 7th May

Open solution

The graph of an object's motion (along the x-axis) is shown in the figure. The instantaneous velocity of the object at points A and B are v_A and v_B respectively. Then



(A) $v_A = v_B = 0.5\ \text{m/s}$

(B) $v_A = 0.5\ \text{m/s} < v_B$

(C) $v_A = 0.5\ \text{m/s} > v_B$

(D) $v_A = v_B = 2\ \text{m/s}$

Q91. AIEEE 2011

Open solution

An object, moving with a speed of 6.25 m/s, is decelerated at a rate given by : $\frac{dv}{dt} = -2.5\sqrt{v}$ where v is the instantaneous speed. The time taken by the object, to come to rest, would be :

(A) 2 s

(B) 4 s

(C) 8 s

(D) 1 s

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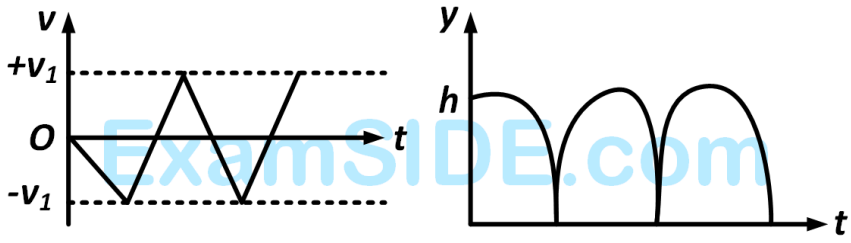
Q92. AIEEE 2009

Consider a rubber ball freely falling from a height $h = 4.9\ m$ onto a horizontal elastic plate. Assume that the duration of collision is negligible and the collision with the plate is totally elastic.

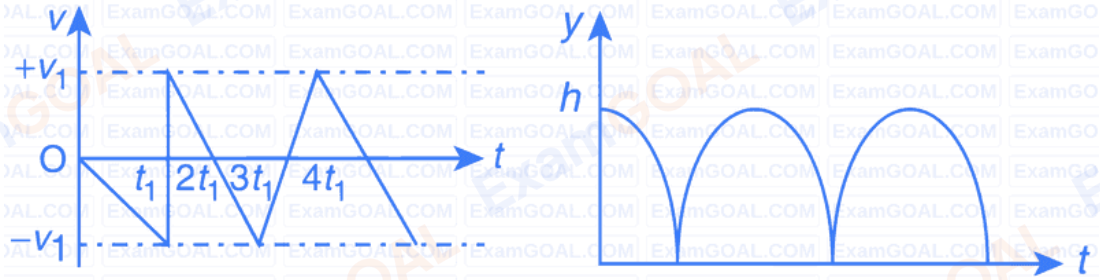
Open solution

Then the velocity as a function of time and the height as a function of time will be :

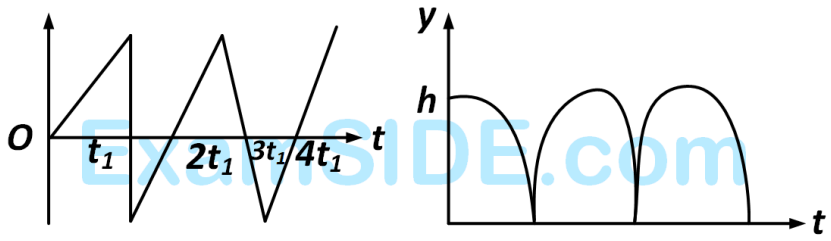
(A)



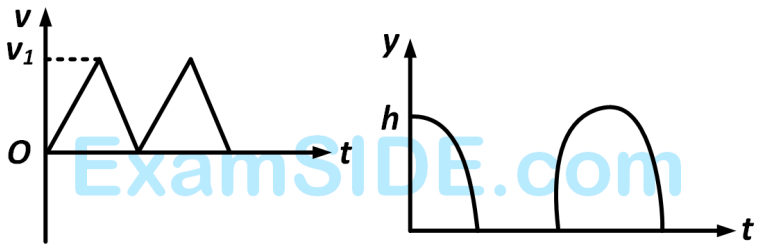
(B)



(C)



(D)



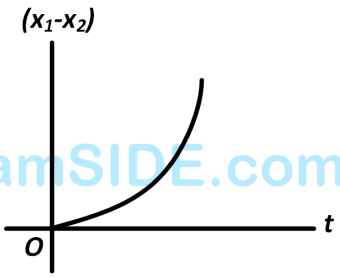
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Q93. AIEEE 2008

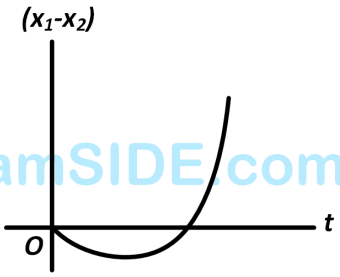
Open solution

A body is at rest at $x = 0$. At $t = 0$, it starts moving in the positive x-direction with a constant acceleration. At the same instant another body passes through $x = 0$ moving in the positive x direction with a constant speed. The position of the first body is given by $x_1(t)$ after time $'t'$; and that of the second body by $x_2(t)$ after the same time interval. Which of the following graphs correctly describes $(x_1 - x_2)$ as a function of time $'t'$?

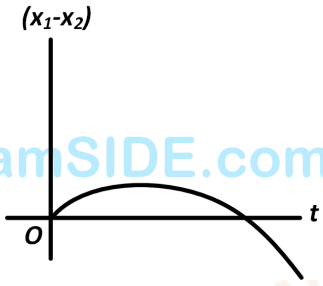
(A)



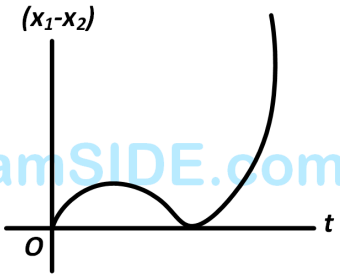
(B)



(C)



(D)



Q94. AIEEE 2007

Open solution

The velocity of a particle is $v = v_0 + gt + ft^2$. If its position is x = 0 at t = 0, then its displacement after unit time (t = 1) is

(A) $v_0 + g/2 + f$ (B) $v_0 + 2g + 3f$ (C) $v_0 + g/2 + f/3$ (D) $v_0 + g + f$

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Q95. AIEEE 2006

A particle located at x = 0 at time t = 0, starts moving along the positive x-direction with a velocity 'v' that varies as $v = \alpha\sqrt{x}$. The displacement of the particle varies with time as

(A) t^2 (B) t (C) $t^{1/2}$ (D) t^3

Open solution

Q96. AIEEE 2005

A car starting from rest accelerates at the rate f through a distance S, then continues at constant speed for time t and then decelerates at the rate $\frac{f}{2}$ to come to rest. If the total distance traversed is 15 S, then

(A) $S = \frac{1}{6}ft^2$ (B) $S = ft$ (C) $S = \frac{1}{4}ft^2$ (D) $S = \frac{1}{72}ft^2$

Open solution

Q97. AIEEE 2005

The relation between time t and distance x is $t = ax^2 + bx$ where a and b are constants. The acceleration is

(A) $2bv^3$ (B) $-2abv^2$ (C) $2av^2$ (D) $-2av^3$

Open solution

Q98. AIEEE 2005

A parachutist after bailing out falls $50\ m$ without friction. When parachute opens, it decelerates at $2\ m/s^2$. He reaches the ground with a speed of $3\ m/s$. At what height, did he bail out?

(A) $182\ m$ (B) $91\ m$ (C) $111\ m$ (D) $293\ m$

Open solution

Q99. AIEEE 2004

A ball is released from the top of a tower of height h meters. It takes T seconds to reach the ground. What is the position of the ball in $\frac{T}{3}$ seconds?

(A) $\frac{8h}{9}$ meters from the ground (B) $\frac{7h}{9}$ meters from the ground
(C) $\frac{h}{9}$ meters from the ground (D) $\frac{7h}{18}$ meters from the ground

Open solution

Q100. AIEEE 2004

An automobile travelling with speed of 60 km/h, can brake to stop within a distance of 20 m. If the car is going twice as fast, i.e 120 km/h, the stopping distance will be

(A) 60 m (B) 40 m (C) 20 m (D) 80 m

Open solution

Q101. AIEEE 2003

A car, moving with a speed of 50 km/hr, can be stopped by brakes after at least 6 m. If the same car is moving at a speed of 100 km/hr, the minimum stopping distance is

(A) 12 m (B) 18 m (C) 24 m (D) 6 m

Open solution

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Q102. AIEEE 2003

The co-ordinates of a moving particle at any time 't' are given by x = αt^3 and y = βt^3. The speed of the particle at time 't' is given by

(A) $3t\sqrt{\alpha^2+\beta^2}$ (B) $3t^2\sqrt{\alpha^2+\beta^2}$

(C) $t^2\sqrt{\alpha^2+\beta^2}$ (D) $\sqrt{\alpha^2+\beta^2}$

Open solution

Q103. AIEEE 2002

From a building two balls A and B are thrown such that A is thrown upwards and B downwards (both vertically with the same speed). If v_A and v_B are their respective velocities on reaching the ground, then

(A) $v_B > v_A$ (B) $v_A = v_B$

(C) $v_A > v_B$ (D) their velocities depend on their masses.

Open solution

Q104. AIEEE 2002

Speeds of two identical cars are u and $4u$ at the specific instant. The ratio of the respective distances in which the two cars are stopped from that instant is :

(A) $1:1$ (B) $1:4$ (C) $1:8$ (D) $1:16$

Open solution

Numerical 26 questions

Q105. JEE Main 2026 (Online) 5th April Morning Shift

From 18 m height above the ground a ball is dropped from rest . The height above the ground at which the magnitude of velocity equal to the magnitude of acceleration (in the same set of units) due to gravity is ____ m.

(Take $g = 10\ \mathrm{m/s^2}$ and neglect the air resistance)

Open solution

Q106. JEE Main 2025 (Online) 2nd April Morning Shift

A person travelling on a straight line moves with a uniform velocity v_1 for a distance x and with a uniform velocity v_2 for the next $\frac{3}{2}x$ distance. The average velocity in this motion is $\frac{50}{7}\ \mathrm{m/s}$. If v_1 is $5\ \mathrm{m/s}$ then $v_2 =$ _____ $\mathrm{m/s}$.

Open solution

Q107. JEE Main 2025 (Online) 29th January Evening Shift

Two cars P and Q are moving on a road in the same direction. Acceleration of car P increases linearly with time whereas car Q moves with a constant acceleration. Both cars cross each other at time t = 0, for the first time. The maximum possible number of crossing(s) (including the crossing at t = 0) is _____.

Open solution

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Q108. JEE Main 2024 (Online) 6th April Evening Shift

A particle moves in a straight line so that its displacement x at any time t is given by $x^2 = 1 + t^2$. Its acceleration at any time t is $x^{-\mathrm{n}}$ where $\mathrm{n} =$ _____.

Open solution

Q109. JEE Main 2024 (Online) 5th April Morning Shift

A body moves on a frictionless plane starting from rest. If $\mathrm{S_n}$ is distance moved between $\mathrm{t} = \mathrm{n} - 1$ and $\mathrm{t} = \mathrm{n}$ and $\mathrm{S_{n-1}}$ is distance moved between $\mathrm{t} = \mathrm{n} - 2$ and $\mathrm{t} = \mathrm{n} - 1$, then the ratio $\frac{\mathrm{S_{n-1}}}{\mathrm{S_n}}$ is $\left(1 - \frac{2}{x}\right)$ for $\mathrm{n} = 10$. The value of x is _____.

Open solution

Q110. JEE Main 2024 (Online) 4th April Evening Shift

A bus moving along a straight highway with speed of $72\ \mathrm{km/h}$ is brought to a halt within $4s$ after applying the brakes. The distance travelled by the bus during this time (Assume the retardation is uniform) is _____ m.

Open solution

Q111. JEE Main 2024 (Online) 1st February Evening Shift

A particle initially at rest starts moving from reference point $x = 0$ along x-axis, with velocity v that varies as $v = 4\sqrt{x}\ \mathrm{m/s}$. The acceleration of the particle is _____ $\mathrm{ms^{-2}}$.

Open solution

Q112. JEE Main 2024 (Online) 1st February Morning Shift

A particle is moving in one dimension (along x axis) under the action of a variable force. Its initial position was $16\ \mathrm{m}$ right of origin. The variation of its position (x) with time (t) is given as $x = -3t^3 + 18t^2 + 16t$, where x is in m and t is in s.

The velocity of the particle when its acceleration becomes zero is _____ $\mathrm{m/s}$.

Open solution

Q113. JEE Main 2024 (Online) 30th January Morning Shift

The displacement and the increase in the velocity of a moving particle in the time interval of t to $(t+1)\mathrm{s}$ are $125\ \mathrm{m}$ and $50\ \mathrm{m/s}$, respectively. The distance travelled by the particle in $(\mathrm{t}+2)^{\mathrm{th}}\mathrm{s}$ is _____ m.

Open solution

Q114. JEE Main 2024 (Online) 27th January Evening Shift

A body falling under gravity covers two points A and B separated by $80\ \mathrm{m}$ in $2\ \mathrm{s}$. The distance of upper point A from the starting point is _____ m (use $\mathrm{g} = 10\ \mathrm{ms^{-2}}$).

Open solution

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Q115. JEE Main 2023 (Online) 1st February Evening Shift

For a train engine moving with speed of $20\ \mathrm{ms}^{-1}$, the driver must apply brakes at a distance of 500 m before the station for the train to come to rest at the station. If the brakes were applied at half of this distance, the train engine would cross the station with speed $\sqrt{x}\ \mathrm{ms}^{-1}$. The value of x is _____.

(Assuming same retardation is produced by brakes)

Open solution

Q116. JEE Main 2023 (Online) 30th January Morning Shift

A horse rider covers half the distance with $5\ \mathrm{m/s}$ speed. The remaining part of the distance was travelled with speed $10\ \mathrm{m/s}$ for half the time and with speed $15\ \mathrm{m/s}$ for other half of the time. The mean speed of the rider averaged over the whole time of motion is $\frac{x}{7}\ \mathrm{m/s}$. The value of x is _____.

Open solution

Q117. JEE Main 2023 (Online) 29th January Morning Shift

A tennis ball is dropped on to the floor from a height of 9.8 m. It rebounds to a height 5.0 m. Ball comes in contact with the floor for 0.2s. The average acceleration during contact is _____ ms^{-2}.

(Given g = 10 ms^{-2})

Open solution

Q118. JEE Main 2022 (Online) 28th July Evening Shift

A ball is thrown vertically upwards with a velocity of $19.6\ \mathrm{ms}^{-1}$ from the top of a tower. The ball strikes the ground after $6\ \mathrm{s}$. The height from the ground up to which the ball can rise will be $\left(\frac{k}{5}\right)\mathrm{m}$. The value of k is _____. (use $\mathrm{g} = 9.8\ \mathrm{m/s}^2$)

Open solution

Q119. JEE Main 2022 (Online) 25th July Evening Shift

A particle is moving in a straight line such that its velocity is increasing at 5 ms^{-1} per meter. The acceleration of the particle is _____ ms^{-2} at a point where its velocity is 20 ms^{-1}.

Open solution

Q120. JEE Main 2022 (Online) 25th July Morning Shift

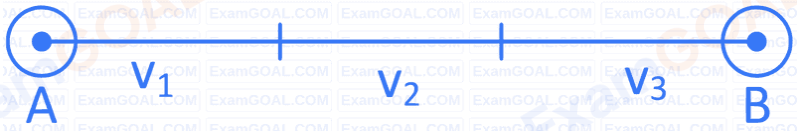
A car is moving with speed of $150\ \mathrm{km/h}$ and after applying the brake it will move $27\ \mathrm{m}$ before it stops. If the same car is moving with a speed of one third the reported speed then it will stop after travelling _____ m distance.

Open solution

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Q121. JEE Main 2022 (Online) 28th June Evening Shift

A car covers AB distance with first one-third at velocity v_1 ms^{-1}, second one-third at v_2 ms^{-1} and last one-third at v_3 ms^{-1}. If $v_3 = 3v_1$, $v_2 = 2v_1$ and $v_1 = 11$ ms^{-1} then the average velocity of the car is _____ ms^{-1}.



Open solution

Q122. JEE Main 2022 (Online) 26th June Evening Shift

A ball is projected vertically upward with an initial velocity of 50 ms^{-1} at t = 0s. At t = 2s, another ball is projected vertically upward with same velocity. At t = _____ s, second ball will meet the first ball (g = 10 ms^{-2}).

Open solution

Q123. JEE Main 2022 (Online) 26th June Morning Shift

A ball of mass 0.5 kg is dropped from the height of 10 m. The height, at which the magnitude of velocity becomes equal to the magnitude of acceleration due to gravity, is _____ m. [Use g = 10 m/s^2]

Open solution

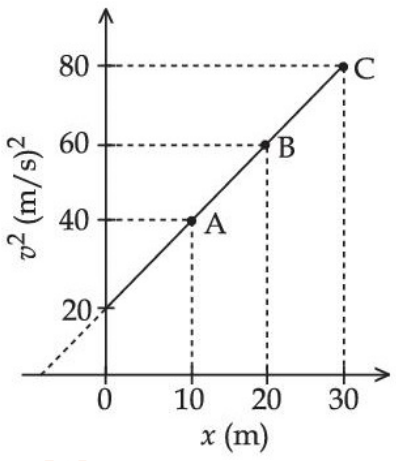
Q124. JEE Main 2022 (Online) 24th June Morning Shift

From the top of a tower, a ball is thrown vertically upward which reaches the ground in 6 s. A second ball thrown vertically downward from the same position with the same speed reaches the ground in 1.5 s. A third ball released, from the rest from the same location, will reach the ground in _____ s.

Open solution

Q125. JEE Main 2021 (Online) 31st August Evening Shift

A particle is moving with constant acceleration 'a'. Following graph shows v^2 versus x(displacement) plot. The acceleration of the particle is _____ m/s^2.



Open solution

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Q126. JEE Main 2021 (Online) 27th August Morning Shift

If the velocity of a body related to displacement x is given by $v = \sqrt{5000 + 24x}$ m/s, then the acceleration of the body is ..... m/s^2.

Open solution

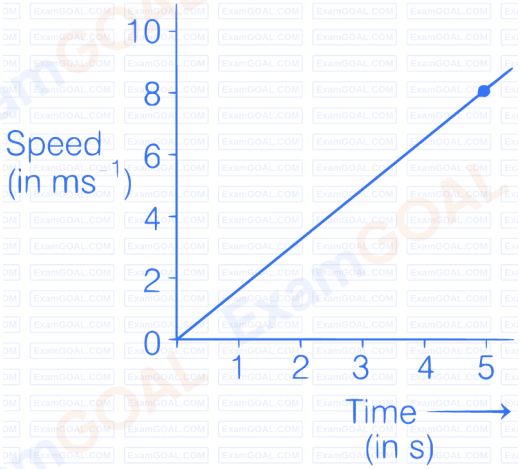
Q127. JEE Main 2021 (Online) 26th August Morning Shift

Two spherical balls having equal masses with radius of 5 cm each are thrown upwards along the same vertical direction at an interval of 3s with the same initial velocity of 35 m/s, then these balls collide at a height of ..... m. (Take g = 10 m/s^2)

Open solution

Q128. JEE Main 2020 (Online) 4th September Evening Slot

The speed versus time graph for a particle is shown in the figure. The distance travelled (in m) by the particle during the time interval t = 0 to t = 5 s will be_____.



Open solution

Q129. JEE Main 2020 (Online) 9th January Morning Slot

The distance x covered by a particle in one dimensional motion varies with time t as $x^2 = at^2 + 2bt + c$. If the acceleration of the particle depends on x as x^{-n}, where n is an integer, the value of n is _____

Open solution

Q130. JEE Main 2020 (Online) 8th January Evening Slot

A ball is dropped from the top of a 100 m high tower on a planet. In the last $\frac{1}{2}s$ before hitting the ground, it covers a distance of 19 m. Acceleration due to gravity (in ms^{-2}) near the surface on that planet is _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **C**	3. **C**	4. **B**	5. **C**
6. **D**	7. **A**	8. **D**	9. **A**	10. **C**
11. **B**	12. **A**	13. **B**	14. **D**	15. **B**
16. **A**	17. **A**	18. **B**	19. **C**	20. **C**
21. **D**	22. **B**	23. **D**	24. **D**	25. **C**
26. **C**	27. **D**	28. **D**	29. **B**	30. **C**
31. **D**	32. **D**	33. **A**	34. **B**	35. **D**
36. **B**	37. **B**	38. **C**	39. **D**	40. **D**
41. **B**	42. **B**	43. **A**	44. **C**	45. **A**
46. **C**	47. **D**	48. **D**	49. **D**	50. **C**
51. **A**	52. **C**	53. **B**	54. **C**	55. **C**
56. **A**	57. **B**	58. **B**	59. **A**	60. **C**
61. **A**	62. **C**	63. **C**	64. **D**	65. **C**
66. **D**	67. **A**	68. **C**	69. **B**	70. **C**
71. **C**	72. **A**	73. **D**	74. **C**	75. **B**
76. **B**	77. **C**	78. **C**	79. **D**	80. **A**
81. **C**	82. **C**	83. **D**	84. **A**	85. **C**
86. **C**	87. **D**	88. **D**	89. **B**	90. **A**
91. **A**	92. **B**	93. **B**	94. **C**	95. **A**
96. **D**	97. **D**	98. **D**	99. **A**	100. **D**
101. **C**	102. **B**	103. **B**	104. **D**	

Numerical

105. **13**	106. **10**	107. **3**	108. **3**	109. **19**
110. **40**	111. **8**	112. **52**	113. **175**	114. **45**
115. **200**	116. **50**	117. **120**	118. **392**	119. **100**
120. **3**	121. **18**	122. **6**	123. **5**	124. **3**
125. **1**	126. **12**	127. **50**	128. **20**	129. **3**
130. **8**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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